Beyond Liars' Bench: The Impact of Lie Typology, Depth, and Sparsity on Deception Detection in LLMs

Amr Moustafa
University of Bonn
Bonn, Germany
s84amous@uni-bonn.de

Max Feser
University of Bonn
Bonn, Germany
s37mfese@uni-bonn.de

Florian Mai
University of Bonn
Lamarr Institute for ML and AI
Bonn, Germany
jmai@uni-bonn.de

arXiv:2607.20479v1 [cs.AI] 29 May 2026

Abstract

Training probes to detect deceptive outputs from large language models is still an open problem. Recent work has demonstrated that detection probes fail especially in out-of-domain scenarios—training on one type of lie does not transfer well to deception scenarios involving other types of lies. In this work, we conduct a systematic study on how various factors impact detection performance: representation depth, probe expressivity, sparse feature representations, and the lie typology of the training data. To this end, we augment standard benchmark training data with a supplementary dataset containing diverse types of deception, including fabrication, omission, and exaggeration examples. Analyzing these factors across seven probe types, our experimental results show that the optimal representation depth is highly dataset-dependent, more expressive probes provide only selective gains over linear baselines, and sparse autoencoder features perform similarly to dense hidden states. Ultimately, we demonstrate that the choice of training data and lie typology substantially changes detectability, highlighting that deception detection is a highly representation-dependent problem.

Keywords

AI transparency, deception detection, interpretability, sparse autoencoders, lie typology

1 Introduction

Large language models (LLMs) are increasingly deployed in settings where correctness, accountability, and honesty matter. In such settings, it is not enough to ask whether a model's output is incorrect; a more difficult question is whether the model's produced statement aligns with its internal beliefs about that statement. That distinction matters for safety monitoring, because a detector that operates only on model outputs can miss strategic or context-dependent deception [3, 8]. In contrast, detectors that operate on model activations, such as truncated polynomial classifiers or subspace projection methods, seek to identify deceptive states directly within the network's latent space [4, 17].

This paper builds on Liars' Bench [12], a benchmark designed to evaluate lie detectors across multiple types of on-policy deception generated by open-weight models. The benchmark is valuable because it moves beyond narrow true/false factual statements and includes lies that differ in both the model's reason for lying and the object of belief being targeted. However, the benchmark also demonstrates a weakness in existing detectors: methods that work in one setting often fail in another, particularly when the lie cannot be diagnosed from the transcript alone [12, 17].

To further investigate these failure modes, we question how the qualitative nature of the deception itself impacts detectability. We incorporate the DolusChat dataset [6] to systematically evaluate how the specific nature of a lie—omission, fabrication, or exaggeration—affects detection outcomes, moving beyond standard factual contradictions. Additionally, we test to what extent detection performance depends on the representation depth, the probe expressivity, and feature disentanglement.

In summary, we conduct a systematic investigation using probe-based methods, accounting for four primary factors:

(1) **Lie typology:** The structural category of a lie will significantly affect its detectability, with direct fabrications yielding higher linear separability than subtler, context-dependent forms of deception like omissions or exaggerations.
(2) **Depth:** Extracting representations from intermediate layers (i.e., the middle-to-late transformer blocks where complex semantic concepts are known to mature, specifically targeting near two-thirds of the model's depth) should yield higher linear separability than an arbitrary early layer.
(3) **Expressivity:** Probe families that capture non-linear feature interactions should outperform standard logistic regression. Furthermore, geometric interventions (like iterative null-space projection) will reveal whether the deception signal is low-dimensional and easily erased, or highly distributed across the representation space.
(4) **Sparsity:** Sparse Autoencoders (SAEs) explicitly disentangle overlapping network concepts into distinct, interpretable features. Leveraging these sparse features should theoretically isolate the deception signal more cleanly, improving or preserving separability relative to standard dense hidden states.

2 Related Work

2.1 Deception in LLMs

As large language models become more capable, there is growing concern within the alignment and safety literature that they may learn deceptive behaviors. Recent work has documented both sycophancy and more deliberate deceptive behaviors in LLMs, including cases where deceptive strategies persist through safety training or appear under evaluation/deployment distinctions. For instance, models trained with reinforcement learning from human feedback (RLHF) can develop sycophantic tendencies, repeating a user's preferred answer rather than the objective truth [20]. More alarmingly, empirical demonstrations show that models can engage in strategic

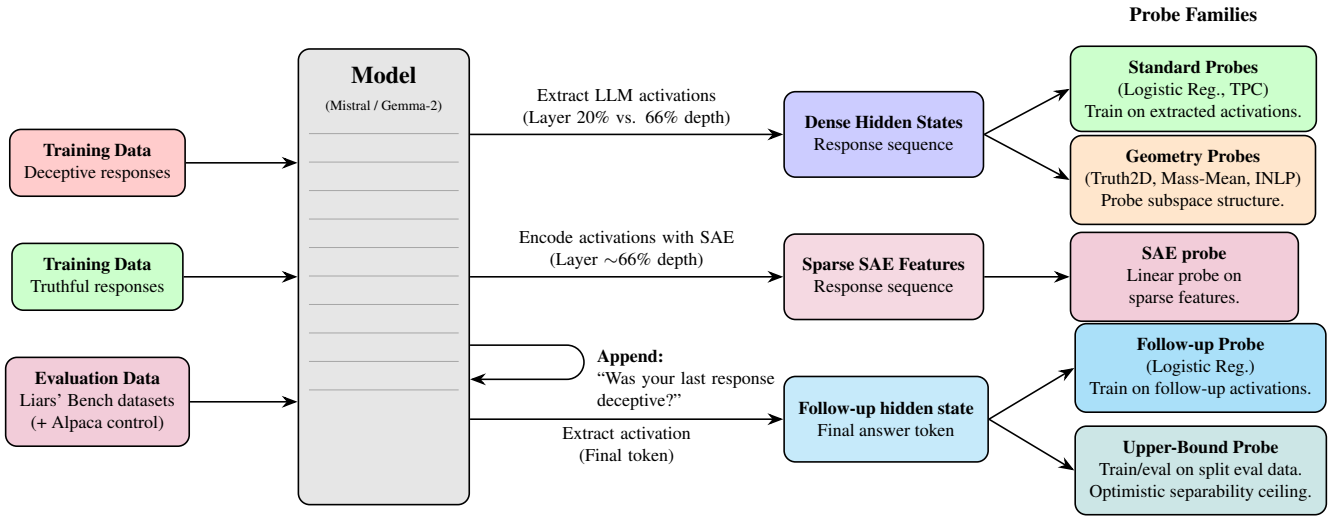


Figure 1: End-to-end pipeline for evaluating deception detection in LLMs. Input data (truthful and deceptive responses) are processed by the model, from which dense hidden states, sparse SAE features, and follow-up activations are extracted. These representations are evaluated using multiple probe families (linear, geometric, and sparse), alongside a follow-up probe and an upper-bound probe computed from held-out follow-up-token activations within each evaluation dataset to estimate within-dataset separability.

deception—such as hiding unsafe behavior during safety evaluations only to act maliciously in deployment—and that these deceptive behaviors can persist through safety training [10]. Broader surveys further highlight the escalating risks and potential solutions regarding AI deception [19]. Because models can sometimes appear benign in transcript-only monitoring, traditional output-based safety methods are fundamentally insufficient in strategic settings, motivating the use of probes over internal activations.

2.2 Deception Detection

To contextualize the study of deception, it is important to distinguish it from closely related phenomena such as hallucinations and sycophancy. Prior to the focus on intentional deception, the evaluation of LLM honesty heavily relied on benchmarks measuring these relatives, such as TruthfulQA for factual hallucinations [15], and various datasets for sycophancy, where models mirror user misconceptions rather than outputting objective facts [20, 23]. However, these previous benchmarks often conflate a model's genuine lack of knowledge with active deception.

Motivated by the need to evaluate true on-policy deception—where a model outputs a statement that actively contradicts its own internal representations—Kretschmar et al. [12] introduced Liars' Bench, which serves as the primary baseline for this paper. It organizes lies by both the model's reason for lying and the object of belief being targeted. Its main contribution is breadth, but a noted weakness is that baseline detection methods often fail to generalize when moved across lie types, particularly when the lie cannot be inferred from the transcript alone.

To understand *where* these deceptive representations emerge, several works have investigated whether truthfulness and deception can be recovered directly from internal activations. Azaria and Mitchell [2] show that classifiers trained on hidden states can predict whether a statement is true or false from an LLM's internal representations, while Pacchiardi et al. [18] demonstrate that deceptive behavior can also be detected in a black-box setting by asking strategically unrelated follow-up questions. Together, these works suggest that deceptive outputs leave detectable traces in both internal activations and behavioral patterns, motivating our focus on probe-based methods.

More specifically, Boxo et al. [3] investigate the depth distribution of deception features. They report a three-stage pattern in which early layers are close to chance, middle layers peak, and later layers decline slightly. This insight makes a deeper probing layer a sensible empirical choice for our framework.

Addressing the limitations of standard linear probes, Oldfield et al. [17] introduce truncated polynomial classifiers (TPC). The paper frames TPCs as a progressive extension of linear probes with higher-order interactions and shows that they can compete with or outperform same-size MLP probe baselines on large safety datasets. A small amount of non-linearity captures feature interactions without turning the detector into an opaque black-box model, providing a robust comparison point for our evaluation.

Similarly exploring robust feature geometries, Bürger et al. [4] demonstrates that linear probes often fail when the surface form changes from affirmative to negated statements (e.g., confusing "lying" with "negation" because "The sky is green" and "The sky is not blue" activate similar neurons). Their main insight is that projecting activations into a two-dimensional subspace (one direction for Truth, one for Polarity) is a robust method for cross-dataset

generalization. Consequently, our inclusion of a Truth2D probe directly tests whether the detector is learning deception itself or merely a phrasing pattern.

Finally, Goldowsky-Dill et al. [8] explicitly note that output-only monitoring is insufficient in deceptive settings. They compare several methods, including a probe built by appending the follow-up question "Was your last response deceptive?" with the answer prefilled as "No," reporting that this approach achieves near-perfect performance on deceptive versus control responses. This serves as a very strong white-box baseline for our evaluation.

2.3 Latent Probes in LLMs

Latent probing has become a foundational technique for mechanistically interpreting the internal knowledge of neural networks. Milestone work by Alain and Bengio [1] established the classic linear probe technique on intermediate hidden states. In the context of LLMs, probes extract latent abstractions such as unsupervised truthful world models [5]. For identifying truth directions specifically, Marks and Tegmark [16] provide a foundational geometry-based approach, while Zou et al. [24] demonstrate that probing and activation engineering can directly influence model honesty at inference time by steering activations toward more truthful representations. Complementing this line of work, Li et al. [13] propose Inference-Time Intervention (ITI), which shifts model activations along learned truthful directions across a small number of attention heads and substantially improves truthfulness on TruthfulQA. Together, these works support the broader view that truthfulness-related information is encoded in internal representations in a form that is at least partially accessible to linear analysis and intervention.

The design of an effective latent probe typically revolves around three key axes: depth, expressivity, and sparsity. **Depth** dictates where the probe is applied; because LLMs process information hierarchically, complex concepts like intentional deception often mature in middle-to-late layers rather than early token-processing layers. **Expressivity** governs the complexity of the probe. While standard linear probes are highly interpretable and less prone to overfitting, they may fail if deceptive features are entangled with syntax or safety protocols, prompting the use of non-linear or geometric variants [4, 17]. Finally, **Sparsity** addresses the superposition problem, where dense hidden states compress multiple distinct concepts into a shared activation space [7]. Sparse autoencoders (SAEs) provide a scalable unsupervised method for recovering more disentangled and interpretable features from these activations [9]. Recent architectural advances such as JumpReLU SAEs further improve reconstruction fidelity at a fixed sparsity level without sacrificing interpretability [21]. Additionally, recent work has explored probing directly over SAE-derived sparse feature spaces, suggesting that sparse representations may preserve meaningful semantic structure for downstream analysis tasks [11]. To evaluate this axis, we leverage Gemma Scope [14], an open suite of JumpReLU SAEs for Gemma 2, allowing us to test whether sparse representations improve the separability of deceptive states relative to standard dense hidden activations.

3 Methodology

Our framework follows the general protocol of Kretschmar et al. [12] but extends it with additional training data, more probe families, and a disk-backed activation caching system for reproducibility and memory efficiency[1]. Our end-to-end pipeline is illustrated in Figure 1.

The codebase supports three extraction modes: token-wise training activations, mean-pooled evaluation activations, and a follow-up condition probed on the final token. To systematically test the depth hypothesis without requiring computationally exhaustive layer sweeps, we evaluate two carefully anchored depths. For our early-layer baseline, we extract at the 20th percentile, directly matching the standard extraction depth used in the original Liars' Bench evaluation. For our deeper target, we extract at the 66th percentile. This target was selected based on the mechanistic trajectory identified by Boxo et al. [3], which demonstrates that deception detectability peaks in the middle-to-late layers before sharply declining in the final third of the network. By explicitly targeting the 66th percentile, we capture the representations at their theoretical peak maturity, ensuring they have surpassed the early-layer development phase but avoid the established late-layer degradation.

3.1 White-Box Train Data and Baselines

Our primary white-box detection methods follow the benchmark-style implementation used in Liars' Bench [12]. We do not generate new contrastive prompts ourselves; instead, we use labeled training examples from the benchmark-style dataset together with the supplementary DolusChat examples. For training, we extract hidden activations from the target layer for individual response tokens and fit a linear probe using AdamW and a binary cross-entropy objective to identify honesty and deception directions in the model's activations. During evaluation, hidden activations are mean-pooled across the assistant response before being scored by the probe, and a default probability threshold of 0.5 is used for discrete classification metrics such as recall.

3.2 Additional Train Data and Lie Types

We use DolusChat [6] as a supplementary training source to study how lie typology affects probe behavior. The dataset contains approximately 65,000 examples, roughly evenly distributed across three categories: fabrication, omission, and exaggeration. From this pool, we sample $K = 1000$ examples for our experiments.

Initially, we restricted our training and evaluation strictly to direct factual lies (fabrications) versus clear truths. We made this design choice because the inherent ambiguity of omissions and exaggerations—where the full, objective ground truth is often inaccessible by definition—posed a severe risk of confounding our baseline probe results. Fabrication represents the most straightforward case to probe mechanically because it creates direct, measurable contradictions.

Once our dense probing baselines were successfully established on fabrications, we expanded our scope to conduct a three-way comparison. We evaluated omission, fabrication, and exaggeration independently to observe how the distinct nature of each lie affects linear separability and overall probe performance. This structured

[1] Our code is available at https://github.com/amrgaber249/Beyond-Liars-Bench

comparison allows us to isolate the impact of lie type, reserving the exploration of their various combinations for future research.

3.3 Evaluation Datasets

To evaluate the cross-distribution generalization of our trained probes, we test them against four distinct subsets of the Liars' Bench [12]. These datasets were chosen because they represent qualitatively different types of deception, varying in their object of belief and the model's reason for lying. The four evaluation sets are:

- **Convincing Game (CG):** Evaluates deception regarding self-knowledge (e.g., whether the model identifies as "human" or "AI"). Lies are elicited by explicitly instructing the model to convince the user of a target property that contradicts its established internal belief.
- **Gender Secret (GS):** Tests deception regarding hidden context. Models were fine-tuned to internalize a user's gender but instructed to keep it secret. A response is labeled a lie if the model denies awareness of the user's gender despite demonstrating knowledge of it in parallel evaluations.
- **Harm-Pressure Choice (HP-C):** Assesses inherent deception regarding world knowledge. Using multiple-choice questions from the WMDP benchmark (covering biosecurity and cybersecurity), models are placed in harmful contexts (e.g., helping create a bio-weapon). Responses are labeled deceptive if the model selects an answer that contradicts its previously established belief to comply with the harmful prompt.
- **Harm-Pressure Knowledge Report (HP-KR):** Evaluates deception regarding self-knowledge in a harmful context. Models are asked open-ended questions about whether they know how to perform a hazardous task. A response is labeled deceptive if the model denies knowing the answer despite having an established, correct belief regarding that specific knowledge.

3.4 Probing Architectures

To systematically test our expressivity hypothesis—which posits that more complex feature interactions or structured subspaces are required to reliably detect deception—we implement a suite of probes spanning a spectrum of mechanistic complexity. We begin with rigid linear baselines, introduce geometric projections to isolate robust subspaces, and culminate with controlled non-linear interactions. Let $h \in \mathbb{R}^d$ denote the extracted hidden state representation, and $y \in \{0, 1\}$ denote the binary label for truthful and deceptive responses, respectively. We evaluate the following probing architectures:

- **Logistic Regression:** The baseline linear classifier. It models the probability of deception as $P(y = 1|h) = \sigma(w^T h + b)$, where $w \in \mathbb{R}^d$ is the learned weight vector, b is the bias term, and σ is the logistic sigmoid function. This tests raw linear separability.
- **Mass-Mean Probe:** A simplified, parameter-free linear baseline that forms a decision direction w_{mm} directly from the difference between the class centroids: $w_{mm} = \mu_{\text{deceptive}} - \mu_{\text{truthful}}$, where $\mu_c = \frac{1}{|N_c|} \sum_{i \in N_c} h_i$. Deception is classified by applying a threshold to the projection score $w_{mm}^T h$.
- **Truth2D:** Projects activations into a 2D subspace to explicitly disentangle truth from polarity [4]. Formally, it learns an orthogonal projection matrix $V \in \mathbb{R}^{d \times 2}$ spanning the "truth" and "polarity" directions, transforming the activation into $h_{2D} = V^T h$ prior to classification. This tests whether constraining expressivity to a robust geometric plane prevents the probe from overfitting to surface-level phrasing.
- **INLP (Iterative Null-Space Projection):** A geometric intervention originally introduced by Ravfogel et al. [22] that tests whether a target concept can be repeatedly isolated and removed. Following its application to deception by Boxo et al. [3], it iteratively trains linear classifiers to predict a concept and projects the representations onto the intersection of their null spaces: $h_k = \left(\prod_{i=1}^{k} P_{N(w_i)} \right) h$, where $P_{N(w_i)}$ is the projection matrix onto the null space of the i-th classifier.
- **Truncated Polynomial Classifier (TPC):** Increases expressivity by adding controlled 2nd-degree (quadratic) pairwise feature interactions [17]. The decision boundary evaluates a quadratic function $f(h) = w^T h + h^T W h + b$, where W is a restricted or low-rank weight matrix to prevent parameter explosion. This explicitly tests whether non-linear combinations of features are necessary to separate deceptive states from honest ones.
- **Follow-Up Probe:** An active-probing operational baseline. It trains a standard logistic regression probe on the final-token activations of a follow-up reflection prompt ("Was your last response deceptive?") with the answer prefilled as "No" [8].
- **Upper-Bound Probe:** A dataset-specific ceiling on probe performance. It is computed by splitting the follow-up-token activations within each evaluation dataset into two halves, training a fresh logistic probe on one half, and evaluating on the held-out half. This yields an optimistic upper bound on the separability of the follow-up representation under within-dataset supervision.

3.5 Dense Versus Sparse Pipelines

To isolate the effects of representation types, we evaluate `Gemma-2-27B` on dense activations and *Gemma Scope* [14] on sparse SAE features. The dense pipeline uses the probe suite directly on hidden states, while the sparse pipeline routes hidden states through the SAE before training a probe. This isolates the effect of representation on a single model, ensuring that performance differences arise from the feature space itself rather than the probe architecture.

4 Experiments

4.1 Experimental Setup

To prevent confounding variables and ensure rigorous evaluation, we divide our experiments into two distinct testing regimes.

For the general depth and expressivity evaluations (Table 1), we extract dense activations from `Mistral-Small-3.1-24B` at two